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consider we have retail db

6 tables

-- we need to find top selling product in each category

-- products and order_items

-- category_id, product_id, product_name,

-- total_quantity, total_amount

let us join the 2 tables

```
select o.order_item_order_id as order_id,
o.order_item_id ,
o.order_item_product_id as product_id,
o.order_item_quantity,
o.order_item_subtotal,
p.product_category_id,
p.product_name
from order_items o
join products p
on o.order_item_product_id = p.product_id
```

-- we need to aggregate the total sale and total quantity

-- sold for each product

```
with products_agg as (select product_category_id,
product_id,product_name,
sum(order_item_quantity) as total_quantity,
sum(order_item_subtotal) as total_amount
from
(select o.order_item_order_id as order_id,
o.order_item_id ,
o.order_item_product_id as product_id,
o.order_item_quantity,
o.order_item_subtotal,
p.product_category_id,
p.product_name
from order_items o
join products p
on o.order_item_product_id = p.product_id) temp
group by product_category_id,product_id,product_name),
windowed_cte as (select *,
row_number() over (partition by product_category_id order by total_amount
desc) as row_num
from products_agg)
select * from windowed_cte where row_num <=1
```

